



BRAC UNIVERSITY

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

Course Code: CSE391

Course Title: Programming for the Internet

PROJECT TITLE

Auto Transport

A Vehicle Shipping Platform: Feature List, Wireframes and Database Schema

Submitted by

Md. Arafat Sarkar

ID: 22101914

1. Feature List

This section records what the Auto Transport platform does at the point of submission. Status is taken from the code base rather than from a plan, so a feature is marked Done only where an implementation exists and is covered by tests. Partial means a working endpoint or table exists without the interface that would make it usable, and Not Built means the supporting schema is in place but no behaviour has been written against it. The platform comprises two clients — a React Native customer application and a staff administration panel — served by a single Laravel API.

1.1 Customer Application

No.	Feature	Status	Notes
1	Register	Done	Name, email, phone, password with confirmation
2	Log in / log out	Done	Sanctum token held in Keychain / Keystore
3	Forgot password	Partial	Issues a real reset link; MAIL_MAILER=log in development, so mail lands in the log
4	Browse transport options	Done	Seven seeded services with pricing and transit times
5	Service detail and public reviews	Done	Rating average, sub-ratings and owner replies
6	Book a shipment	Done	Addresses, date, up to eight vehicles, operable toggle
7	Switch transport option mid-form	Done	Re-prices live without discarding the form
8	My shipments	Done	Status badge, lane, price and outstanding balance
9	Track a shipment	Done	Customer-visible timeline only
10	Cancel a shipment	Done	Refused by the state machine once delivered
11	Leave a review after delivery	Done	Overall rating plus four optional sub-ratings
12	Mark a review helpful	Done	One vote per user; an author cannot vote on their own review
13	Contact form	Done	Throttled; lands in the staff inbox

No.	Feature	Status	Notes
14	Profile and address book	Done	Partial updates supported; email address is not editable
15	Account status and roles	Done	Displays an unverified-email warning
16	Instant price estimate	Partial	Endpoint exists and is tested; no dedicated screen
17	Guest quote request	Partial	Endpoint accepts guests; no screen, as the app books directly
18	Accept or decline a staff quote	Not Built	Declared in endpoints.ts; no controller behind it
19	Push notifications	Not Built	device_tokens table and endpoint exist; nothing dispatches
20	Live map tracking	Not Built	booking_events carries latitude and longitude; no map screen

Table 1.1: Customer-facing features (20 items).

1.2 Staff Administration Panel

No.	Feature	Status	Notes
21	Staff login	Done	Five failures trigger a 15-minute lock; six requests per minute per IP
22	Dashboard	Done	Seven statistic tiles, each gated individually by permission
23	User management	Done	Create, edit, suspend, soft-delete and assign roles
24	Role and permission matrix	Done	Six roles and fourteen permissions, editable without a deploy
25	Review moderation	Done	Approve, reject with reason, feature and reply
26	Dispatch board	Done	Filter by status and pickup window; keyword search
27	Booking detail and status moves	Done	Only legal state transitions are offered
28	Quote request queue	Done	Read-only intake with versioned quote history

No.	Feature	Status	Notes
29	Contact inbox	Done	Opening a message marks it read; replies are recorded
30	Service catalogue editing	Done	Pricing in cents, visibility flag and sort order
31	Settings	Done	Typed values; encrypted secrets are never re-rendered
32	Activity log	Partial	Writes on service, role and user changes; no browsing interface
33	Issue a quote / re-price	Not Built	Schema supports versioning; no form
34	Assign carrier, driver or truck	Not Built	Columns and models exist; no interface
35	Email delivery of replies	Not Built	Audit trail only, and the form states as much

Table 1.2: Staff-facing features (15 items).

1.3 Platform and Cross-Cutting Concerns

No.	Feature	Status	Notes
36	Two-layer authorisation	Done	Coarse staff gate, then per-resource permissions
37	ULID public identifiers	Done	Integer primary keys never leave the server
38	Integer-cents money	Done	A { cents, currency } pair end to end
39	Booking state machine	Done	Status and timeline event written in one transaction
40	Denormalised rating aggregates	Done	Observer-maintained, with reviews:rebuild-aggregates as a repair path
41	Append-only booking timeline	Done	Internal notes are filtered out of the API response
42	Address snapshotting	Done	A booking keeps a record of where it actually went
43	Payment ledger schema	Partial	Tables and model exist; no gateway wired
44	Distance lookup	Partial	Stubbed with a 1000-mile fallback
45	SEO metadata	Partial	Polymorphic seo_meta table; no public web frontend

No.	Feature	Status	Notes
46	Guest quote claiming on verification	Not Built	Must gate on email_verified_at

Table 1.3: Platform-level features (11 items).

1.4 Summary

Across the three groups the platform records 46 features: 30 built, 9 partial and 7 not built. The partial and unbuilt items cluster in two areas — the quoting pipeline and the carrier assignment workflow — both of which have complete database support and await interface work rather than modelling.

2. Wireframes

The wireframes below describe the two clients. They are drawn at low fidelity so that structure, state and navigation are legible without committing to a visual treatment. Every screen shown corresponds to a route that exists in the application.

2.1 Navigation Map

The customer application is organised around a four-item tab bar. Authentication sits outside the tab bar, and the review form is reachable only from a shipment that has reached the delivered state — a constraint enforced by the API rather than by the client.

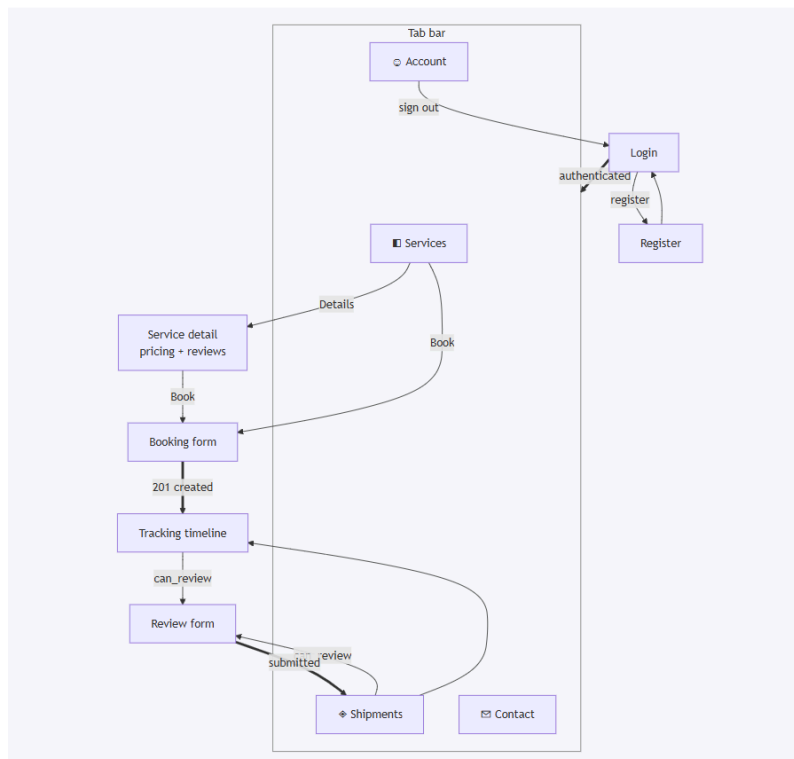


Figure 2.1: Customer application navigation map, showing the tab bar boundary and the delivery-gated review path.

2.2 Customer Application Screens

The service card is the repeating unit of the catalogue: rating, price floor, transit window and two actions. Its shape is inherited by the detail screen and by the shipment list further down the flow.

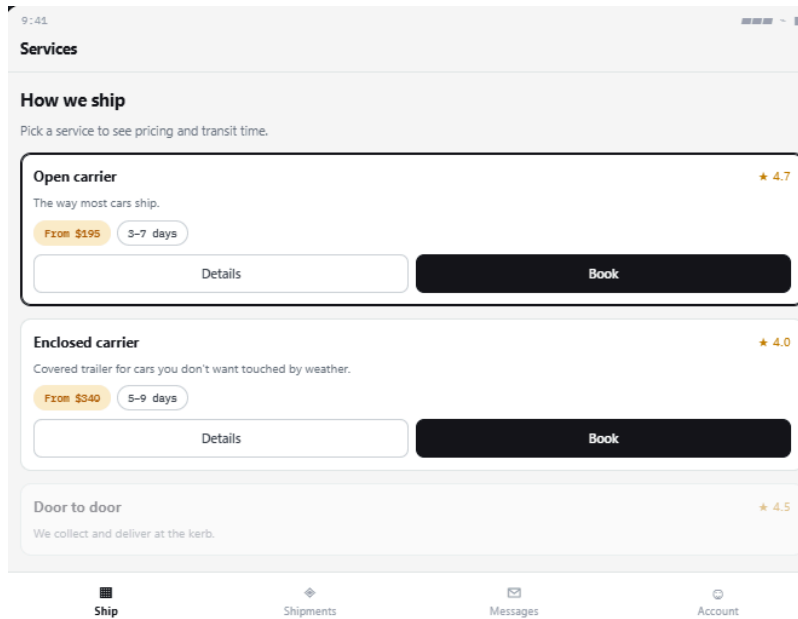


Figure 2.2: Services list. Cards carry a rating, an indicative price and a transit window.

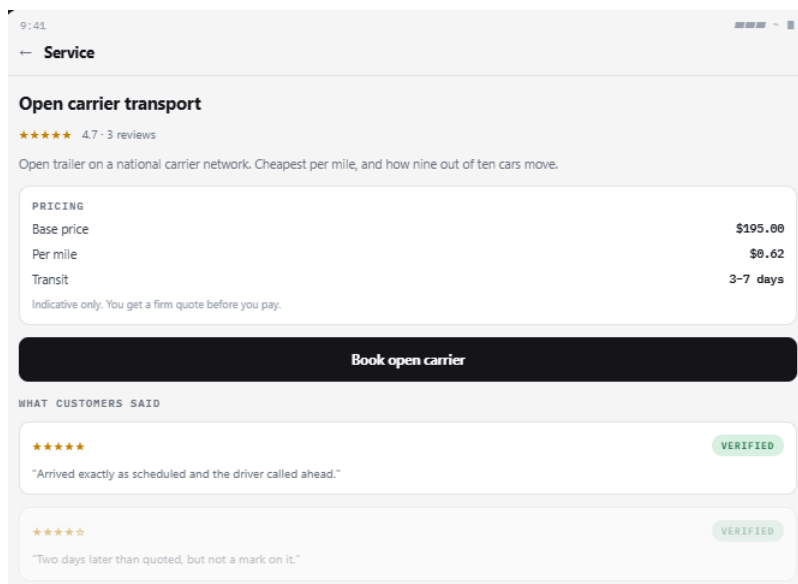


Figure 2.3: Service detail. Pricing model above, moderated customer reviews below.

9:41

← Book a shipment

Book a shipment

Open carrier — from \$195

TRANSPORT OPTION

Open Enclosed Door

Typically 3–7 days

COLLECTION

Street address

City State

ZIP

DELIVERY

Street address

City State

ZIP

VEHICLES 1 OF 8

Sedan SUV Moto

Year Make Model

It starts and drives

A non-runner needs a winch, which adds to the price.

+ Add another vehicle

Confirm booking

The price is an automated estimate and can change after the carrier inspects the vehicle.

Figure 2.4: Booking form. Four stacked sections and a single commit action; the vehicle section repeats up to eight times.

9:41

Shipments

Your shipments

BK-2026-000002

Austin → Denver

Picked up

2026-08-01

\$1,240.00

Track shipment

Leave a review

DELIVERED

BK-2026-000003

Austin → Denver

Picked up

2026-08-14

\$1,180.00

Track shipment

IN TRANSIT

Ship Shipments Messages Account

Figure 2.5: Shipment list. The status badge determines which actions the card offers.

9:41

← Shipment

BK-2026-000002 DELIVERED

Austin → Denver

SCHEDULE

Pickup 2026-08-01

Delivery 2026-08-06

PAYMENT

Total \$1,240.00

Balance due \$0.00

TRACKING

- Booked 01 Aug
- Picked up 02 Aug
- In transit 03 Aug
- Delivered 06 Aug

Figure 2.6: Shipment detail. Schedule and payment summaries above an append-only tracking timeline.

9:41

← Leave a review

Austin → Denver

BK-2026-000002 - delivered 06 Aug

OVERALL RATING

★★★★★

RATE THE DETAILS

Communication ★★★★★

Timeliness ★★★★★

Vehicle condition ★★★★★

Value ★★★★★

HEADLINE

Sum it up in one line

YOUR REVIEW

Tell other shippers how it went

Submit review

Reviews are checked before they appear publicly.

Figure 2.7: Review form. One overall rating and four optional sub-ratings, submitted for moderation.

2.3 Staff Administration Panel

The administration panel places every resource behind one persistent shell: a grouped sidebar, a page title bar and a two-level breadcrumb. Only the login screen renders outside this shell.

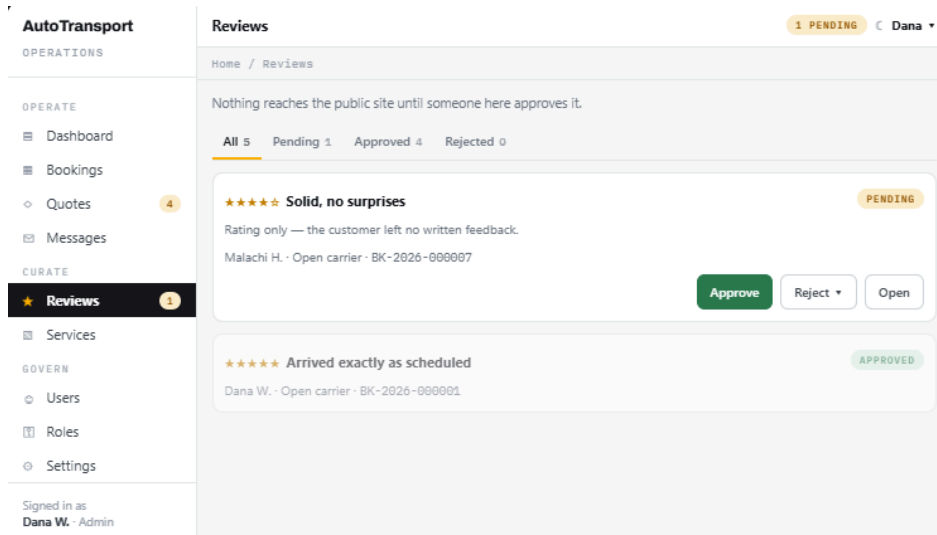


Figure 2.8: Review moderation queue. Approve and reject are available on the row, so the common case never requires opening the record.

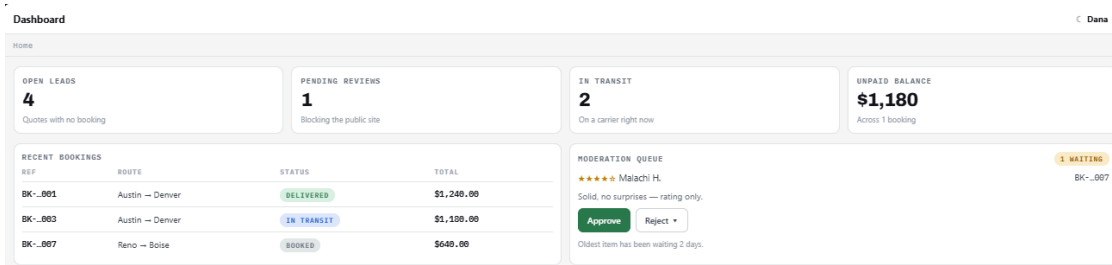


Figure 2.9: Staff dashboard. Four counters and two queues, each linking into a filtered list.

3. Schema

The database comprises 40 tables. This section presents the inventory, the core transactional relationships, the state vocabularies that govern them and the permission set used for authorisation. Design rationale — in particular why quote requests and quotes are modelled as two tables, why addresses are snapshotted onto a booking, and why money is stored as integer cents — is documented separately in the database design document.

3.1 Table Inventory

Context	Tables
Identity and access	users, roles, permissions, role_has_permissions, model_has_roles, model_has_permissions, addresses, device_tokens, personal_access_tokens, password_reset_tokens, sessions
Catalogue	service_categories, services, vehicle_types, locations
Supply	carriers, driver_profiles, trucks
Sales pipeline	quote_requests, quote_request_vehicles, quotes
Fulfilment	bookings, booking_vehicles, booking_events
Money	payments, payment_webhooks
Reputation	reviews, review_votes
Content and SEO	pages, seo_meta, faqs, contact_messages, settings
Cross-cutting	media, activity_log, jobs, job_batches, failed_jobs, cache, cache_locks

Table 3.1: The 40 tables grouped by the concern they serve.

3.2 Core Transaction Flow

A quote request is submitted by a user or by a guest, is priced by one or more versioned quotes, and converts into a booking. The booking owns its vehicles, its event timeline and its payments, and may be reviewed exactly once after delivery. Ratings are denormalised onto the service record and maintained by an observer.

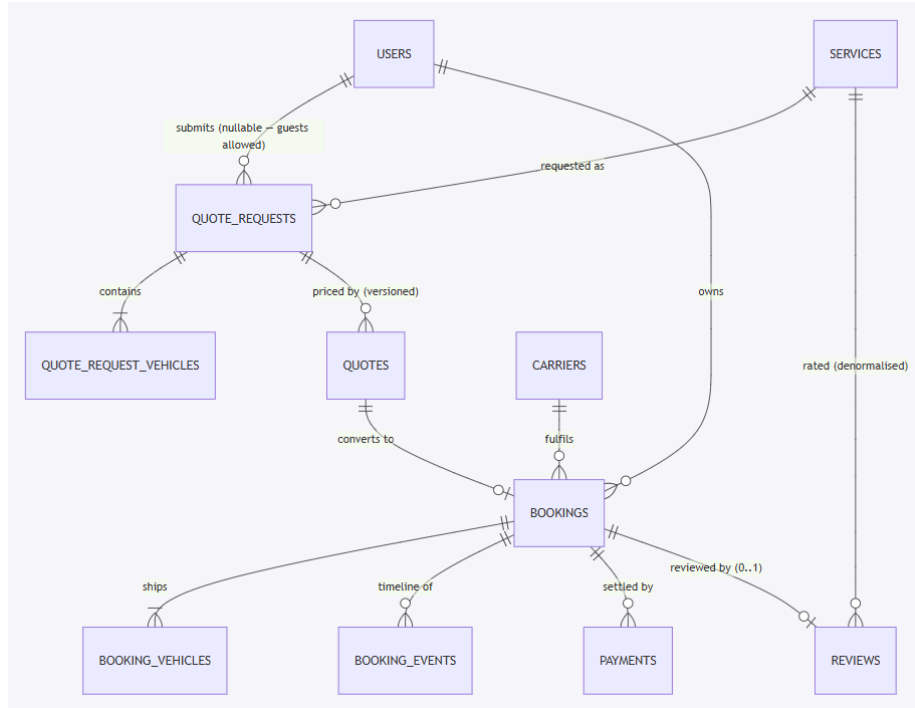


Figure 3.1: Entity relationships across the core transaction flow.

3.3 State Vocabularies

Enumeration	Values
BookingStatus	pending_payment, confirmed, assigned, picked_up, in_transit, delivered, cancelled
QuoteRequestStatus	new, reviewing, quoted, accepted, declined, expired, spam
QuoteStatus	draft, sent, viewed, accepted, declined, expired, superseded
PaymentStatus	pending, authorized, captured, failed, refunded, disputed
ReviewStatus	pending, approved, rejected
UserRole	super-admin, admin, dispatcher, support, driver, customer

Table 3.2: The six enumerations that define legal record states.

Each of these is stored as a VARCHAR column paired with a PHP backed enum, rather than as a native MySQL ENUM. Adding a state is therefore a code change instead of a locking ALTER TABLE that must be coordinated with a deploy.

3.4 Permissions

Authorisation is applied in two layers: a coarse gate that admits staff to the panel at all, and a fine-grained permission check on each resource. Six roles draw from the following fourteen permissions, and the mapping between them is editable at runtime.

view_bookings	manage_bookings
view_quotes	manage_quotes
view_reviews	moderate_reviews
view_contact_messages	manage_contact_messages
view_users	manage_users
manage_carriers	manage_content
manage_settings	view_reports

Table 3.3: The fourteen permissions available to the role matrix.